

Assignment 1

Advanced Machine Learning

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Exercise 1

1. **(0.3 points)** What is the maximum value of the natural even number n , $n = 2m$, such that there exists a hypothesis class $\mathcal{H}$ with n elements that shatters a set C of $m = \frac{n}{2}$ points? Give an example of such an $\mathcal{H}$ and C . Justify your answer.

To shatter a set C of size m , the hypothesis class $\mathcal{H}$ must be large enough to realize every possible case over C . This requires:

$$|\mathcal{H}| \geq 2^m$$

Given that $n = 2^m$, the condition becomes $2m \geq 2^m$ for $m \in \mathbb{N}^*$. Checking small values:

For $m = 1$: $2(1) \geq 2^1 \Rightarrow 2 \geq 2$ ✓

For $m = 2$: $2(2) \geq 2^2 \Rightarrow 4 \geq 4$ ✓

For $m = 3$: $2(3) \geq 2^3 \Rightarrow 6 \geq 8$ ✗

We claim $m = 2$ is the largest integer satisfying the inequality. Define:

$$g : [3, +\infty) \rightarrow \mathbb{R}, \quad g(x) = 2x - 2^x$$

Computing the derivative gives:

$$g'(x) = 2 - 2^x \ln 2$$

Since $2^x \ln 2 > 2$ for all $x \geq 3$, we have $g'(x) < 0$ throughout $[3, +\infty)$, so g is strictly decreasing on this interval. This means:

$$g(x) \leq g(3) = 6 - 8 = -2 < 0, \quad \forall x \in [3, +\infty)$$

This confirms that $2m < 2^m$ holds for every integer $m \geq 3$, making $m = 2$ (or $n = 4$) the maximum valid value.

Take $C = \{x_1, x_2\}$ and define $\mathcal{H} = \{h_1, h_2, h_3, h_4\}$ via the following labeling table:

	x_1	x_2
h_1	0	0
h_2	1	0
h_3	0	1
h_4	1	1

Every one of the $2^2 = 4$ distinct binary labelings of C appears as a row, so $\mathcal{H}$ shatters C .

Exercise 2

2. (1.2 points) Consider $\mathcal{H} = \mathcal{H}_1 \cup \mathcal{H}_2 \cup \mathcal{H}_3$, where:

$$\mathcal{H}_1 = \{h_a : \mathbb{R} \rightarrow \{0, 1\} \mid h_a(x) = \mathbf{1}_{[x \geq a]}(x) = \mathbf{1}_{[a, +\infty)}(x), a \in \mathbb{R}\},$$

$$\mathcal{H}_2 = \{h_a : \mathbb{R} \rightarrow \{0, 1\} \mid h_a(x) = \mathbf{1}_{[x < a]}(x) = \mathbf{1}_{(-\infty, a)}(x), a \in \mathbb{R}\},$$

$$\mathcal{H}_3 = \{h_{a,b} : \mathbb{R} \rightarrow \{0, 1\} \mid h_{a,b}(x) = \mathbf{1}_{[a \leq x \leq b]}(x) = \mathbf{1}_{[a,b]}(x), a, b \in \mathbb{R}\}.$$

Consider the realizability assumption.

a) Compute $\text{VCdim}(\mathcal{H})$.

b) Give an algorithm A and determine an upper bound on the sample complexity $m_{\mathcal{H}}(\epsilon, \delta)$ such that the definition of PAC-learnability is satisfied.

(a)

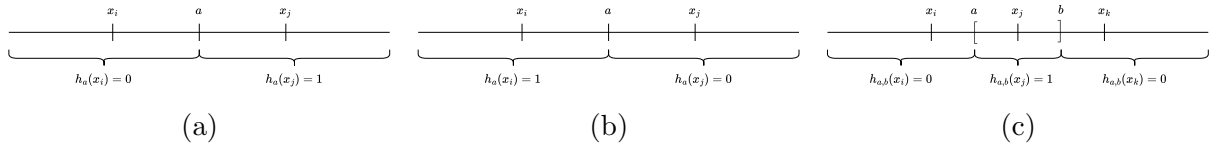


Figure 1

We will compute the VCdim for each hypothesis class.

$\mathcal{H}_1$: Consider two points $x_1, x_2 \in \mathbb{R}$ with $x_1 < x_2$. We observe the following edge-case:

(1, 0): This would require $a \leq x_1$ and $a > x_2$ simultaneously, which is impossible since $x_1 < x_2$, see Figure 1a.

Since the labeling (1, 0) cannot be realized, no set of size 2 is shattered. However, any single point can be shattered, so $\text{VCdim}(\mathcal{H}_1) = 1$.

$\mathcal{H}_2$: Consider two points $x_1, x_2 \in \mathbb{R}$ with $x_1 < x_2$. We observe the following edge-case:

(0, 1): This would require $a > x_1$ and $a \leq x_2$ simultaneously, which is impossible since $x_1 < x_2$, see Figure 1b.

Since the labeling (0, 1) cannot be realized, no set of size 2 is shattered. However, any single point can be shattered, so $\text{VCdim}(\mathcal{H}_2) = 1$.

$\mathcal{H}_3$: For any two points $x_1 < x_2$, all four labelings are achievable via appropriate interval $[a, b]$, so $\mathcal{H}_3$ shatters any set of size 2. Moving to three points $x_1 < x_2 < x_3$, the labeling (1, 0, 1) turns out to be unrealizable: achieving it would force $x_1, x_3 \in [a, b]$ while $x_2 \notin [a, b]$, but since $x_1 < x_2 < x_3$, having both endpoints inside $[a, b]$ forces the entire segment $[x_1, x_3] \subseteq [a, b]$, which necessarily includes x_2 , see Figure 1c. Thus, no set of size 3 is shattered, giving $\text{VCdim}(\mathcal{H}_3) = 2$.

$\mathcal{H}$: Since $\mathcal{H}_3 \subseteq \mathcal{H}$, any two points $x_1 < x_2$ are shattered by $\mathcal{H}$. It remains to check whether any three points $x_1 < x_2 < x_3$ can be shattered. The labeling (1, 0, 1) is the critical case:

In $\mathcal{H}_1$: realizing $(1, 0, 1)$ requires $a \leq x_1$ and $a > x_2$, a **contradiction**.

In $\mathcal{H}_2$: realizing $(1, 0, 1)$ requires $a > x_3$ and $a \leq x_2$, a **contradiction**.

In $\mathcal{H}_3$: realizing $(1, 0, 1)$ requires $x_1, x_3 \in [a, b]$ while $x_2 \notin [a, b]$, which is impossible as argued above.

Since $(1, 0, 1)$ is not realizable by any hypothesis in $\mathcal{H}$, no set of three points is shattered, and therefore $\text{VCdim}(\mathcal{H}) = 2$.

(b)

Since we are under the realizability assumption, we can use the *Empirical Risk Minimization (ERM)* algorithm. Let $S = \{(x_i, y_i)\}_{i=1}^m$ be the training sample. The algorithm A returns any hypothesis $h \in \mathcal{H}$ such that $L_S(h) = 0$. Specifically:

1. Check if there exists $h \in \mathcal{H}_1$ consistent with S . If so, return it.
2. Otherwise, check if there exists $h \in \mathcal{H}_2$ consistent with S . If so, return it.
3. Otherwise, since the problem is realizable, there must exist an interval $h_{a,b} \in \mathcal{H}_3$ consistent with S . A specific choice is the smallest consistent interval:

$$a = \min\{x_i : y_i = 1\}, \quad b = \max\{x_i : y_i = 1\}$$

For a hypothesis class with $\text{VCdim}(\mathcal{H}) = d$, the sample complexity for PAC learnability in the realizable case is bounded by:

$$m_{\mathcal{H}}(\epsilon, \delta) \leq \frac{C}{\epsilon} \left(d \log \frac{1}{\epsilon} + \log \frac{1}{\delta} \right)$$

where C is a constant (course 7, slide 32/36). Using the specific upper bound provided by the Fundamental Theorem of Statistical Learning for the realizable case, see Theorem 28.3[1]:

$$m_{\mathcal{H}}(\epsilon, \delta) \leq \frac{16d \log(16e/\epsilon) + 8 \log(2/\delta)}{\epsilon}$$

Substituting the result from part (a), where $d = 2$:

$$m_{\mathcal{H}}(\epsilon, \delta) \leq \frac{32 \log(16e/\epsilon) + 8 \log(2/\delta)}{\epsilon}$$

Thus, $m_{\mathcal{H}}(\epsilon, \delta) = O\left(\frac{1}{\epsilon} \left(\log \frac{1}{\epsilon} + \log \frac{1}{\delta}\right)\right)$.

Exercise 3

3. (1.2 points) Usually, there is a correlation between the number of parameters of a hypothesis class $\mathcal{H}$ and its VC-dimension. Consider the following hypothesis classes:

$$\mathcal{H}_1 = \{h_a : \mathbb{R} \rightarrow \{0, 1\} \mid h_a(x) = \mathbf{1}_{[a, a+1]}(x), a \in \mathbb{R}\},$$

$$\mathcal{H}_2 = \{h_b : \mathbb{R} \rightarrow \{0, 1\} \mid h_b(x) = \mathbf{1}_{[b, b+1] \cup [b+2, b+4]}(x), b \in \mathbb{R}\},$$

$$\mathcal{H}_3 = \{h_c : \mathbb{R} \rightarrow \{0, 1\} \mid h_c(x) = \mathbf{1}_{[c, c+1] \cup [c+2, c+4] \cup [c+6, c+9]}(x), c \in \mathbb{R}\}.$$

Consider the realizability assumption. Compute $\text{VCdim}(\mathcal{H}_1)$, $\text{VCdim}(\mathcal{H}_2)$, $\text{VCdim}(\mathcal{H}_3)$.

$\text{VCdim}(\mathcal{H}_1)$

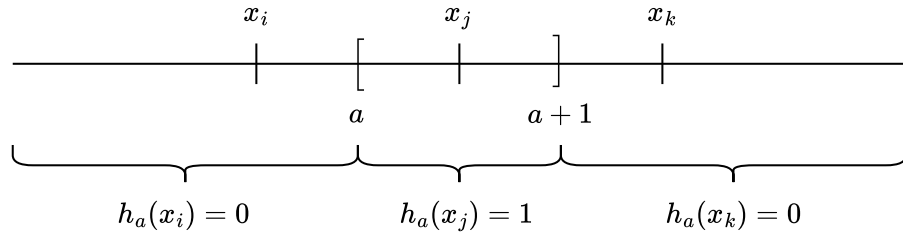


Figure 2

For 2 points: all labelings are achievable.

For 3 points: we will show that the labeling $(1, 0, 1)$ is **impossible**.

Let $x_1 < x_2 < x_3$, which implies:

$$x_2 \in [x_1, x_3] \tag{1}$$

The labeling $(1, 0, 1)$ requires:

$$h_a(x_1) = 1, \quad h_a(x_2) = 0, \quad h_a(x_3) = 1$$

However:

$$\left. \begin{array}{l} h_a(x_1) = h_a(x_3) = 1 \implies [x_1, x_3] \subseteq [a, a+1] \\ (1) \implies x_2 \in [x_1, x_3] \end{array} \right\} \implies x_2 \in [a, a+1] \implies h_a(x_2) = 1 \quad \text{(contradiction)}$$

Therefore, $\text{VCdim}(\mathcal{H}_1) = 2$.

$\text{VCDim}(\mathcal{H}_2)$

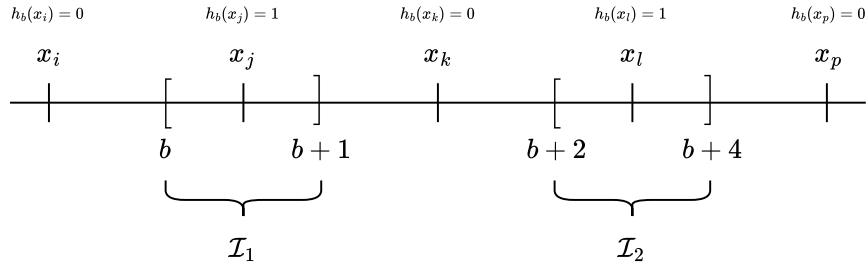


Figure 3

For 2 points: all labelings are achievable.

For 3 points: all labelings are achievable.

For 4 points:

Let $x_1 < x_2 < x_3 < x_4$

Consider the labeling: $(1, 0, 0, 1)$

$$\left. \begin{array}{l} h_b(x_1) = h_b(x_4) = 1 \\ h_b(x_2) = h_b(x_3) = 0 \end{array} \right\} \Rightarrow x_1 \in \mathcal{I}_1 \text{ and } x_4 \in \mathcal{I}_2$$

$$\Rightarrow x_2, x_3 \in (b+1, b+2)$$

$$\Rightarrow x_3 - x_1 < 2$$

Consider the labeling: $(1, 0, 1, 0)$

$$\left. \begin{array}{l} h_b(x_1) = 1 \\ h_b(x_3) = 1 \\ x_3 - x_1 < 2 \end{array} \right\} \Rightarrow x_1, x_3 \in \mathcal{I}_1 \text{ or } x_1, x_3 \in \mathcal{I}_2$$

$$\Rightarrow [x_1, x_3] \subseteq \mathcal{I}_1 \text{ or } [x_1, x_3] \subseteq \mathcal{I}_2$$

However:

$$x_2 \in [x_1, x_3] \Rightarrow x_2 \in \mathcal{I}_1 \text{ or } x_2 \in \mathcal{I}_2 \Rightarrow h_b(x_2) = 1 \quad (\text{contradiction})$$

Therefore, $\text{VCdim}(\mathcal{H}_2) = 3$.

$\text{VCDim}(\mathcal{H}_3)$

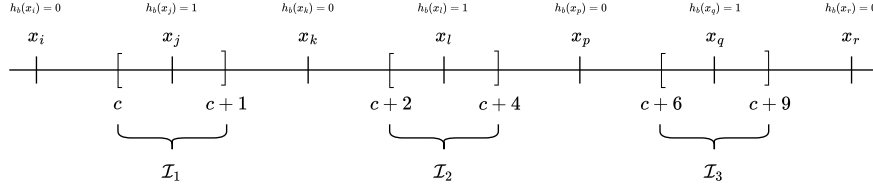


Figure 4

For 2 points: all labelings are achievable.

For 3 points: all labelings are achievable.

For 4 points: all labelings are achievable.

For 5 points:

Let $x_1 < x_2 < x_3 < x_4 < x_5$

Consider the labeling $(1, 0, 1, 0, 1)$

$$\left. \begin{array}{l} x_1 \in \mathcal{I}_1, x_3 \in \mathcal{I}_2, x_5 \in \mathcal{I}_3 \\ x_2 \in (c+1, c+2) \\ x_4 \in (c+4, c+6) \end{array} \right\} \implies x_3 - x_1 > 1 \text{ and } x_5 - x_3 > 2 \implies \quad (2)$$

$$\implies x_5 - x_1 > 3$$

Consider the labeling $(1, 0, 0, 0, 1)$. Note that x_1 and x_5 cannot lie in the same interval (2).

Case 1: $x_1 \in \mathcal{I}_1, x_5 \in \mathcal{I}_2$

$$\begin{aligned} x_1 \in \mathcal{I}_1, x_5 \in \mathcal{I}_2 &\implies x_2, x_3, x_4 \in (c+1, c+2) \implies \\ &\implies x_4 - x_2 < 1, x_4 - x_3 < 1, x_3 - x_2 < 1 \end{aligned}$$

Case 2: $x_1 \in \mathcal{I}_2, x_5 \in \mathcal{I}_3$

$$\begin{aligned} x_1 \in \mathcal{I}_2, x_5 \in \mathcal{I}_3 &\implies x_2, x_3, x_4 \in (c+4, c+6) \implies \\ &\implies x_4 - x_2 < 2, x_4 - x_3 < 2, x_3 - x_2 < 2 \end{aligned}$$

Case 3: $x_1 \in \mathcal{I}_1, x_5 \in \mathcal{I}_3$

$$x_1 \in \mathcal{I}_1, x_5 \in \mathcal{I}_3 \implies x_2, x_3, x_4 \in (c+1, c+2) \text{ or } x_2, x_3, x_4 \in (c+4, c+6)$$

From the Pigeonhole Principle, there exist at least 2 points from $\{x_2, x_3, x_4\}$ that both lie in either $(c+1, c+2)$ or $(c+4, c+6)$. This implies:

$$\begin{aligned} \exists i \in \{2, 4\} \text{ s.t. } x_3, x_i \text{ both lie in either } (c+1, c+2) \text{ or } (c+4, c+6) &\implies \\ \implies |x_3 - x_i| < 2 \end{aligned}$$

However, for the labeling $(1, 0, 1, 0, 1)$, it is impossible for $x_3 \in \mathcal{I}_2$ with $h_c(x_2) = h_c(x_4) = 0$ and $x_3 - x_2 < 2$ and $x_4 - x_3 < 2$.

Therefore, $\text{VCdim}(\mathcal{H}_3) = 4$.

References

- [1] Shai Shalev-Shwartz and Shai Ben-David. *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press, 2014, p. 398. URL: <http://www.cs.huji.ac.il/~shais/UnderstandingMachineLearning>.